FORMELSAMLING FÖR ELEKTROMAGNETISM.

Maxwells ekvationer:

$$\overline{\nabla} \cdot \overline{D} = \rho \,, \qquad \overline{\nabla} \times \overline{E} = -\frac{\partial \overline{B}}{\partial t} \,, \qquad \overline{\nabla} \cdot \overline{B} = 0 \,, \qquad \overline{\nabla} \times \overline{H} = \overline{J} + \frac{\partial \overline{D}}{\partial t}$$

$$\overline{D} \equiv \varepsilon_0 \overline{E} + \overline{P} \,, \qquad \rho_p \equiv -\overline{\nabla} \cdot \overline{P} \,, \qquad \rho_{sp} \equiv \overline{P} \cdot \hat{n} \,, \qquad \text{Poyntingvektor: } \overline{\mathcal{P}} \equiv \overline{E} \times \overline{H}$$

$$\overline{B} \equiv \mu_0 \left(\overline{H} + \overline{M} \right) \,, \qquad \overline{J}_m \equiv \overline{\nabla} \times \overline{M} \,, \qquad \overline{J}_{sm} \equiv \overline{M} \times \hat{n}$$

Elektrisk potential, V, och elektrisk fältstyrka, $\overline{E}$, från elektriskt dipolmoment.

$$V = \frac{p \, \cos\theta}{4\pi\varepsilon_0 \, r^2} \,, \qquad \overline{E} = \frac{p}{4\pi\varepsilon_0 \, r^3} \left(2 \, \cos\theta \, \hat{r} + \sin\theta \, \hat{\theta} \right)$$

Magnetisk vektorpotential, $\overline{A}$, och magnetisk flödestäthet, $\overline{B}$, från magnetiskt dipolmoment.

$$\overline{A} = \frac{\mu_0 \, \overline{m} \times \hat{r}}{4\pi r^2} \,, \qquad \overline{B} = \frac{\mu_0 \, m}{4\pi r^3} \left(2 \, \cos\theta \, \hat{r} + \sin\theta \, \hat{\theta} \right)$$

Kraftmoment: $\overline{T} = \overline{r} \times \overline{F} \,, \qquad \overline{T} = \overline{m} \times \overline{B}$

Elektromotorisk spänning: $\varepsilon_{\text{ems}} = \oint_C \left(\overline{v} \times \overline{B} \right) \cdot \overline{\mathrm{d}\ell} + \int_S \left(-\frac{\partial \overline{B}}{\partial t} \right) \cdot \overline{\mathrm{d}S} = -\frac{\mathrm{d}\Phi}{\mathrm{d}t}$

Några vanliga integraler:

$$\int \frac{\mathrm{d}x}{(x^2+a^2)^{1/2}} = \ln(x+(x^2+a^2)^{1/2}) \qquad \int \frac{\mathrm{d}x}{(x^2+a^2)^{3/2}} = \frac{x}{a^2(x^2+a^2)^{1/2}}$$

$$\int \frac{x^2\,\mathrm{d}x}{(x^2+a^2)^{1/2}} = \frac{x}{2}(x^2+a^2)^{1/2} - \frac{a^2}{2}\ln(x+(x^2+a^2)^{1/2}) \qquad \int \frac{\mathrm{d}x}{x^2+a^2} = \frac{1}{a}\arctan(\frac{x}{a})$$

$$\int \frac{x^2\,\mathrm{d}x}{(x^2+a^2)^{3/2}} = \frac{-x}{(x^2+a^2)^{1/2}} + \ln(x+(x^2+a^2)^{1/2}) \qquad \int \frac{x^2\,\mathrm{d}x}{x^2+a^2} = x - a\,\arctan(\frac{x}{a})$$

$$\int \frac{x^3\,\mathrm{d}x}{(x^2+a^2)^{3/2}} = 2\,(x^2+a^2)^{1/2} - \frac{x^2}{(x^2+a^2)^{1/2}}$$

Några användbara vektoridentiteter (V och f är godtyckliga skalära funktioner, $\overline{A}$, $\overline{B}$ och $\overline{C}$ är godtyckliga vektorvärda funktioner.):

$$\overline{A}\cdot(\overline{B}\times\overline{C}) = \overline{B}\cdot(\overline{C}\times\overline{A}) = \overline{C}\cdot(\overline{A}\times\overline{B}) \qquad \overline{A}\times(\overline{B}\times\overline{C}) = \overline{B}\,(\overline{A}\cdot\overline{C}) - \overline{C}\,(\overline{A}\cdot\overline{B})$$

$$\overline{\nabla}(fV) = f\,\overline{\nabla}V + V\,\overline{\nabla}f \qquad \overline{\nabla}\cdot(f\overline{A}) = f\,\overline{\nabla}\cdot\overline{A} + \overline{A}\cdot\overline{\nabla}f$$

$$\overline{\nabla}\times(f\,\overline{A}) = f\,\overline{\nabla}\times\overline{A} + \overline{\nabla}f\times\overline{A} \qquad \overline{\nabla}\cdot(\overline{A}\times\overline{B}) = \overline{B}\cdot(\overline{\nabla}\times\overline{A}) - \overline{A}\cdot(\overline{\nabla}\times\overline{B})$$

$$\overline{\nabla}\cdot(\overline{\nabla}V) = \overline{\nabla}^2V \qquad \overline{\nabla}\times(\overline{\nabla}V) = \overline{0}$$

$$\overline{\nabla}\times(\overline{\nabla}\times\overline{A}) = \overline{\nabla}(\overline{\nabla}\cdot\overline{A}) - \overline{\nabla}^2\overline{A} \qquad \overline{\nabla}\cdot(\overline{\nabla}\times\overline{A}) = 0$$

Gradient, divergens, rotation och Laplace-operatorn i tre olika koordinatsystem.
(V är en godtycklig skalär funktion och $\overline{A}$ är en godtycklig vektorvärd funktion).

<u>Cartesiska koordinater</u> (x,y,z):

$$\overline{\nabla}V = \frac{\partial V}{\partial x}\,\hat{x} + \frac{\partial V}{\partial y}\,\hat{y} + \frac{\partial V}{\partial z}\,\hat{z}$$

$$\overline{\nabla}\cdot\overline{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\overline{\nabla}\times\overline{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right]\hat{x} + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right]\hat{y} + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right]\hat{z}$$

$$\overline{\nabla}^2V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Cylindriska koordinater (R,ϕ,z):

$$\overline{\nabla} V = \frac{\partial V}{\partial R}\,\hat{R} + \frac{1}{R}\,\frac{\partial V}{\partial \phi}\,\hat{\phi} + \frac{\partial V}{\partial z}\,\hat{z}$$

$$\overline{\nabla} \cdot \overline{A} = \frac{1}{R}\,\frac{\partial}{\partial R}\,(R\,A_R) + \frac{1}{R}\,\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\overline{\nabla} \times \overline{A} = \left[\frac{1}{R}\,\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right]\hat{R} + \left[\frac{\partial A_R}{\partial z} - \frac{\partial A_z}{\partial R}\right]\hat{\phi} + \frac{1}{R}\,\left[\frac{\partial}{\partial R}\,(R\,A_\phi) - \frac{\partial A_R}{\partial \phi}\right]\hat{z}$$

$$\overline{\nabla}^2 V = \frac{1}{R}\,\frac{\partial}{\partial R}\left(R\,\frac{\partial V}{\partial R}\right) + \frac{1}{R^2}\,\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Sfäriska koordinater (r,θ,ϕ):

$$\overline{\nabla} V = \frac{\partial V}{\partial r}\,\hat{r} + \frac{1}{r}\,\frac{\partial V}{\partial \theta}\,\hat{\theta} + \frac{1}{r\,\sin\theta}\,\frac{\partial V}{\partial \phi}\,\hat{\phi}$$

$$\overline{\nabla} \cdot \overline{A} = \frac{1}{r^2}\,\frac{\partial}{\partial r}\left(r^2\,A_r\right) + \frac{1}{r\,\sin\theta}\,\frac{\partial}{\partial \theta}\,(\sin\theta\,A_\theta) + \frac{1}{r\,\sin\theta}\,\frac{\partial A_\phi}{\partial \phi}$$

$$\begin{aligned}\overline{\nabla} \times \overline{A} \;\; = \;\; & \frac{1}{r\,\sin\theta}\,\left[\frac{\partial}{\partial \theta}\,(\sin\theta\,A_\phi) - \frac{\partial A_\theta}{\partial \phi}\right]\hat{r} + \frac{1}{r}\,\left[\frac{1}{\sin\theta}\,\frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r}\,(r\,A_\phi)\right]\hat{\theta} + \\ & \frac{1}{r}\,\left[\frac{\partial}{\partial r}\,(r\,A_\theta) - \frac{\partial A_r}{\partial \theta}\right]\hat{\phi}\end{aligned}$$

$$\overline{\nabla}^2 V = \frac{1}{r^2}\,\frac{\partial}{\partial r}\left(r^2\,\frac{\partial V}{\partial r}\right) + \frac{1}{r^2\,\sin\theta}\,\frac{\partial}{\partial \theta}\left(\sin\theta\,\frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2\,\sin^2\theta}\,\frac{\partial^2 V}{\partial \phi^2}$$